

ASG Airlines:End-to-End Data Engineering Case Study

Documentation Report by,

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Problem Statement

ASG Airlines, a nationwide carrier, operates flights across multiple cities through its booking platforms, scheduling systems, and airport logs. The organization collects operational flight data from these various systems; however, inconsistencies in the dataset such as corrupted flight identifiers, inconsistent time formats, missing values, and unhandled overnight (cross-day) flight scenarios have created challenges in generating accurate operational insights.

Without a standardized and reliable data pipeline, the airline faces difficulties in calculating accurate flight durations, analyzing route-wise traffic, identifying delays and anomalies, and making data-driven operational decisions.

To improve operational efficiency and reporting accuracy, ASG Airlines aims to build an end-to-end data engineering pipeline that ingests, cleans, standardizes, and transforms flight data into a reliable analytical dataset for reporting and business intelligence purposes.

Chapter 1

Introduction

1.1 Background

1.1.1 Introduction to Airline Operational Data

The airline industry generates a large amount of operational data through its day-to-day activities. Flight schedules, departure and arrival times, flight numbers, routes, aircraft operations, and airport information are recorded for each flight. These records are useful for monitoring airline performance, identifying operational problems, and supporting decisions related to scheduling and resource utilization.

However, airline data is often collected from different operational systems and may not follow a common format. As a result, datasets may contain missing values, inconsistent identifiers, duplicate records, incorrect time values, or unnecessary columns. If these issues are not addressed before analysis, they can affect the accuracy of the results and lead to incorrect conclusions.

For this reason, proper data processing is an important part of airline operational analytics. The data must be collected, cleaned, standardized, and organized before it can be used for reporting and decision-making.

1.1.2 Importance of Data Engineering in Airline Operations

Data engineering provides the processes required to convert raw operational data into a structured and usable form. Airline datasets can originate from different sources such as flight management systems, airport systems, booking platforms, and operational logs. These sources may use different formats and naming conventions.

A well-designed data pipeline helps bring these datasets into a common structure. It can also identify data quality issues and apply appropriate transformations before the data is made available for analysis.

In this project, data engineering is used to handle issues such as missing airline information, irregular flight identifiers, duplicate records, unnecessary columns, and incorrect calculations for flights that cross midnight. The processed data is then stored in different layers so that each stage of transformation can be maintained and analyzed separately.

1.1.3 Role of Analytics in Flight Scheduling and Operational Monitoring

Once flight data has been cleaned and structured, analytics can be used to understand operational performance. Route-level and airline-level analysis can provide useful information about flight duration, traffic patterns, departure periods, and unusual records.

The analysis can help operations teams to:

- Compare the number of flights operated by different airlines and routes.
- Identify routes with unusually high or low flight durations.
- Examine flight activity during different periods of the day.
- Identify routes where duration values differ significantly from the normal pattern.
- Understand the distribution of flights between major airports.
- Monitor operational indicators through interactive dashboards.

Such analysis provides a clearer view of the dataset and can support further investigation of operational issues.

1.1.4 Problems Caused by Inconsistent and Incomplete Operational Data

The quality of the input data has a direct effect on the quality of the analysis. In the ASG Airlines dataset, several issues were identified during the initial examination of the data.

Some records contained missing or unknown airline information. Flight identifiers also contained prefixes that could be used to identify the corresponding airline, but this information was not directly available as a separate field.

The raw dataset also contained additional columns generated during the data ingestion process. These columns did not contain useful operational information and could result in unnecessary schema complexity.

Another important issue was the calculation of flight duration. For flights that departed before midnight and arrived after midnight, directly subtracting the arrival time from the departure time produced an incorrect negative value.

These issues demonstrate the need for a systematic data cleaning and transformation process before performing operational analysis.

1.2 Project Overview

1.2.1 Introduction to the ASG Airlines Case Study

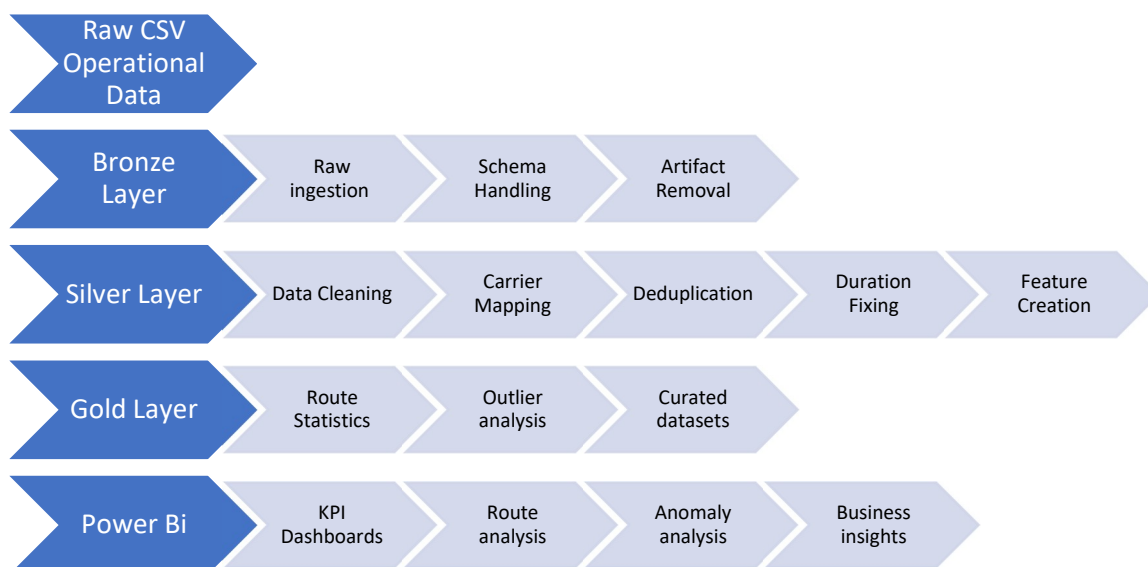
This project is based on a case study of **ASG Airlines**, a regional airline network operating domestic flights through multiple carriers. The dataset contains flight records associated with four major airlines: **IndiGo, Air India, SpiceJet, and Vistara**.

During the initial examination of the operational data, several data quality problems were identified. These included missing airline values, irregular flight identifiers, unnecessary columns generated during ingestion, duplicate records, and incorrect duration values for flights crossing midnight.

The project therefore focuses on developing a complete data engineering and business intelligence workflow to clean the raw dataset, organize the processed information, perform statistical analysis, and present the results through Power BI dashboards.

1.2.2 Overview of the End-to-End Data Pipeline

The project follows a **Medallion Architecture**, consisting of Bronze, Silver, and Gold layers. Each layer represents a different stage of data processing.



The major technologies used in the pipeline are described below.

Databricks and PySpark:

Databricks was used as the main data processing environment. PySpark was used to read the operational data, perform transformations, handle missing values, extract airline information, calculate flight durations, and generate additional analytical fields.

Delta Lake:

Delta tables were used to maintain the different stages of the processed data. The Bronze, Silver, and Gold layers provide a structured approach to maintaining the raw, cleaned, and analytical versions of the dataset.

Power BI:

Power BI was used to create the final analytical dashboards. The processed Gold data was used to develop KPI measures, route analysis, duration analysis, anomaly analysis, and other visualizations required for operational monitoring.

1.3 Problem Statement

The ASG Airlines operational dataset contained several data quality and analytical issues that affected its direct use for reporting. The major problems identified were:

1. **Missing Airline Information:**
Some flight records contained null or unknown airline values, making it difficult to group and compare flights by carrier.
2. **Inconsistent Flight Identifiers:**
Flight numbers contained different formats and prefixes. These prefixes contained useful information about the airline but had not been separated into a dedicated field.
3. **Unnecessary Ingestion Columns:**
The raw data contained additional columns such as _c7, _c8, _c9, and _c10. These columns were not required for the analysis and increased the complexity of the dataset.
4. **Incorrect Duration for Overnight Flights:**
Flights crossing midnight produced incorrect duration values when the arrival time was numerically smaller than the departure time. For example, a flight departing at 23:30 and arriving at 01:30 would result in an incorrect negative duration if the date change was not considered.
5. **Lack of Route-Based Anomaly Detection:**
A single duration threshold cannot be applied equally to every route because different routes naturally have different average flight durations. A route-specific statistical baseline was therefore required.
6. **Limited Operational Reporting:**
The raw dataset did not provide an integrated dashboard through which users could easily examine flight volume, duration patterns, route performance, and unusual records.

The project addresses these issues by implementing a structured data engineering pipeline followed by statistical analysis and business intelligence reporting.

1.4 Aim of the Project

The main aim of this project is to **design and implement an end-to-end data engineering and business intelligence pipeline for cleaning, transforming, analyzing, and visualizing airline operational data.**

The project processes raw flight records using Databricks and PySpark, stores the transformed data through a Bronze-Silver-Gold architecture, applies route-level statistical analysis to identify unusual flight durations, and presents the resulting operational information through an interactive Power BI dashboard.

1.5 Objectives

The major objectives of the project are:

- To ingest the raw airline operational dataset into a structured Bronze layer.
- To identify and remove unnecessary columns and ingestion artifacts.
- To clean and standardize the operational records using PySpark.
- To identify missing airline information from flight number prefixes where possible.
- To remove duplicate records from the dataset.
- To correctly calculate flight duration for flights crossing midnight.
- To create additional features such as route and departure time-of-day.

- To calculate route-level average and standard deviation of flight duration.
- To identify unusual flight durations using a route-specific statistical threshold.
- To prepare curated datasets suitable for business intelligence analysis.
- To develop a Power BI dashboard for monitoring airline and route-level operations.
- To provide useful insights regarding flight volume, duration patterns, routes, and operational anomalies.

1.6 Scope of the Project

1.6.1 Included in the Project Scope

The project covers the processing and analysis of **1,005 flight records** representing **four major airlines**: IndiGo, Air India, SpiceJet, and Vistara.

The dataset covers **30 unique domestic routes** and includes information required for flight-level and route-level analysis.

The following activities are included in the project:

- Raw CSV data ingestion using Databricks.
- Data processing using PySpark.
- Bronze, Silver, and Gold data layers.
- Removal of unnecessary ingestion columns.
- Missing airline identification and mapping.
- Duplicate record handling.
- Flight duration correction for overnight flights.
- Route and time-of-day feature creation.
- Route-level statistical analysis.
- Duration anomaly identification using the $\mu + 1.5\sigma$ threshold.
- Preparation of analytical datasets for Power BI.
- Development of interactive Power BI dashboards.
- KPI development using DAX.
- Route, airline, duration, and anomaly analysis.

The project also considers data governance concepts such as role-based access control, row-level security, and protection of sensitive information as possible considerations for a production environment.

The project primarily focuses on **batch data processing, data quality improvement, descriptive analysis, statistical anomaly detection, and business intelligence reporting**.

Chapter Summary

This chapter introduced the background and motivation for applying data engineering techniques to airline operational data. The major data quality problems in the ASG Airlines dataset were identified, including missing carrier information, inconsistent flight identifiers, ingestion artifacts, and incorrect duration calculations for overnight flights.

The proposed solution uses a Bronze-Silver-Gold architecture implemented with Databricks, PySpark, and Delta Lake. The processed data is then used to perform route-level analysis and develop interactive Power BI dashboards. The following chapters describe the methodology, data processing steps, implementation, analysis, and results of the project in greater detail.

PROBLEM STATEMENT AND OBJECTIVES

2.1 Problem Definition

Airline operations generate a large amount of data related to flight numbers, airlines, departure and arrival times, routes, and airport operations. This information can be used to monitor flight activity and identify unusual operational patterns. However, the usefulness of such analysis depends on the quality and consistency of the underlying data.

The ASG Airlines dataset used in this project contains domestic flight records collected from operational sources. During the initial examination of the dataset, several issues were identified, including missing airline information, inconsistent flight identifiers, unnecessary columns, duplicate records, and incorrect duration calculations for flights crossing midnight.

These problems make it difficult to perform reliable route-level and airline-level analysis directly on the raw data. Therefore, a structured data engineering pipeline is required to clean and transform the records before they are used for statistical analysis and business intelligence reporting.

2.2 Operational and Data Issues in Flight Records

The examination of the raw dataset identified the following major issues.

2.2.1 Unnecessary Columns from Data Ingestion

The raw dataset contained additional columns such as `_c7`, `_c8`, `_c9`, and `_c10`. These columns did not contain useful information for the analysis and appeared as a result of the structure of the source data during ingestion.

Keeping these columns can make the dataset more difficult to manage and may introduce unnecessary schema-related issues during subsequent processing. Therefore, these fields were removed during the initial data preparation stage.

2.2.2 Missing Airline Information

Some records contained null or unknown values in the airline column. This creates a problem when comparing flight volumes and operational characteristics across airlines.

The `flight_id` field contains airline-related prefixes in the available records. These prefixes can therefore be used to identify the corresponding airline and recover missing airline information where the prefix can be reliably extracted.

The mapping used in the project is:

Flight Prefix Airline

6F	IndiGo
AI	Air India
SJ	SpiceJet
UK	Vistara

2.2.3 Inconsistent Flight Identifiers

The `flight_id` field contains alphanumeric values that include information about the operating airline. However, this information was not available as a separate attribute in the raw dataset.

For example, prefixes such as 6F, AI, SJ, and UK can be extracted from the flight identifier and mapped to their respective airlines. Separating this information makes the dataset more consistent and supports airline-level analysis.

Regular expression-based extraction was used during the data-cleaning process to identify the relevant prefix patterns.

2.2.4 Incorrect Duration for Flights Crossing Midnight

Flight duration cannot always be calculated by directly subtracting the arrival time from the departure time because some flights cross midnight.

First, the departure and arrival times are converted into minutes from midnight:

$$T_{dep} = 60H_{dep} + M_{dep}$$

$$T_{arr} = 60H_{arr} + M_{arr}$$

where:

- H_{dep} = departure hour
- M_{dep} = departure minute
- H_{arr} = arrival hour
- M_{arr} = arrival minute
- T_{dep} = departure time expressed in minutes
- T_{arr} = arrival time expressed in minutes

The flight duration D is then calculated as:

$$D = \begin{cases} T_{arr} - T_{dep}, & \text{if } T_{arr} \geq T_{dep} \\ (T_{arr} + 1440) - T_{dep}, & \text{if } T_{arr} < T_{dep} \end{cases}$$

Here, **1,440 minutes represents 24 hours**.

For example, consider a flight departing at **23:15** and arriving at **01:45**.

$$T_{dep} = (23 \times 60) + 15 = 1395$$

$$T_{arr} = (1 \times 60) + 45 = 105$$

Since:

$$T_{arr} < T_{dep}$$

the second condition is applied:

$$D = (105 + 1440) - 1395$$

$$D = 150 \text{ minutes}$$

Therefore, the calculated duration is **150 minutes**, or **2 hours and 30 minutes**.

This correction prevents negative duration values and ensures that flights crossing midnight are represented correctly in the analytical dataset.

2.2.5 Lack of Route-Specific Outlier Detection

Flight durations naturally vary between different routes. A single duration threshold applied to all routes may therefore incorrectly classify normal flights as anomalies or fail to identify unusual flights on shorter routes.

To address this issue, the project calculates the mean and standard deviation of flight duration separately for each route.

For a given route r , the mean duration is:

$$\mu_r = \frac{1}{n_r} \sum_{i=1}^{n_r} D_i$$

where:

- D_i = duration of flight i
- n_r = number of flights on route r
- μ_r = mean duration for route r

The route-level standard deviation is calculated as:

$$\sigma_r = \sqrt{\frac{1}{n_r} \sum_{i=1}^{n_r} (D_i - \mu_r)^2}$$

The anomaly threshold for each route is then defined as:

$$Threshold_r = \mu_r + 1.5\sigma_r$$

A flight is classified as an anomaly when:

$$D_i > Threshold_r$$

This route-specific approach provides a local reference point for identifying unusually long flight durations rather than applying the same threshold to every route.

2.2.6 Limited Business Intelligence Reporting

The raw dataset does not provide an integrated interface for examining the results of the analysis. Information such as flight volume, route distribution, duration patterns, and anomalous records must be derived through separate analysis steps.

To address this limitation, the cleaned and enriched data is connected to Power BI, where interactive dashboards are developed for operational analysis and reporting.

2.3 Aim of the Project

The main aim of this project is to **design and implement an end-to-end data engineering and business intelligence pipeline for ASG Airlines operational flight data.**

The project uses Azure Databricks, PySpark, and Delta Lake to process the raw dataset through Bronze, Silver, and Gold layers. The cleaned data is then analyzed using route-level statistical measures and presented through an interactive Power BI dashboard.

2.4 Technical Objectives

The project has the following objectives.

2.4.1 Raw Data Ingestion

To ingest the raw CSV flight dataset into the Bronze layer using Databricks and store the data in Delta format for subsequent processing.

2.4.2 Data Cleaning and Standardization

To identify and correct data quality problems such as missing values, duplicate records, unnecessary columns, and inconsistent data formats.

2.4.3 Airline Identification

To extract airline prefixes from the flight_id field and use them to fill missing airline values where sufficient information is available.

The mappings used are:

Flight Prefix Airline

6F	IndiGo
AI	Air India
SJ	SpiceJet
UK	Vistara

2.4.4 Cross-Midnight Duration Correction

To develop a duration calculation method that correctly handles flights where the arrival time occurs after midnight.

The duration is calculated using:

$$D = \begin{cases} T_{arr} - T_{dep}, & \text{if } T_{arr} \geq T_{dep} \\ (T_{arr} + 1440) - T_{dep}, & \text{if } T_{arr} < T_{dep} \end{cases}$$

This ensures that overnight flights produce valid positive duration values.

2.4.5 Feature Engineering

To create additional fields that support operational analysis, including:

- Route (source_airport → destination_airport)
- Departure hour
- Time-of-day category
- Corrected flight duration
- Route-level mean duration
- Route-level standard deviation
- Anomaly classification

Departure time is grouped into four categories:

- **Morning**
- **Afternoon**
- **Evening**
- **Night**

2.4.6 Route-Level Statistical Analysis

To calculate route-specific mean duration and standard deviation and use these values to identify unusually long flight records.

For each route r :

$$\mu_r = \frac{1}{n_r} \sum_{i=1}^{n_r} D_i$$
$$\sigma_r = \sqrt{\frac{1}{n_r} \sum_{i=1}^{n_r} (D_i - \mu_r)^2}$$

The route-specific threshold is:

$$Threshold_r = \mu_r + 1.5\sigma_r$$

A flight is flagged when:

$$D_i > Threshold_r$$

2.4.7 Gold Layer Preparation

To create curated datasets in the Gold layer containing the cleaned and enriched information required for reporting and analysis.

2.4.8 Power BI Dashboard Development

To develop a multi-page Power BI dashboard that presents the processed data through KPIs and interactive visualizations.

The dashboard covers:

- Executive overview

- Flight duration analysis
- Route and hub analysis
- Anomaly analysis
- Key influencer analysis

2.5 Project Scope and Boundary Conditions

2.5.1 Included Within the Scope

The project includes the following areas:

- Processing of **1,005 flight records**.
- Analysis of **4 airlines**: IndiGo, Air India, SpiceJet, and Vistara.
- Analysis of **30 unique domestic routes**.
- Data ingestion and processing using **Azure Databricks**.
- Data transformation using **PySpark**.
- Data storage using **Delta Lake**.
- Implementation of Bronze, Silver, and Gold processing layers.
- Data cleaning and standardization.
- Airline identification using flight number prefixes.
- Duplicate record handling.
- Cross-midnight duration correction.
- Route and time-of-day feature creation.
- Route-level statistical analysis.
- Duration anomaly identification.
- Power BI data modeling and visualization.
- Development of interactive operational dashboards.

The analysis also covers major airports represented in the dataset, including:

- Mumbai (BOM)
- Delhi (DEL)
- Kolkata (CCU)
- Bangalore (BLR)
- Chennai (MAA)
- Hyderabad (HYD)

2.5.2 Duration and Statistical Boundaries

The project analyzes flight durations within the observed dataset range of approximately **30 to 300 minutes**.

For anomaly detection, route-level mean and standard deviation are calculated from the available records. The threshold is defined as:

$$Threshold_r = \mu_r + 1.5\sigma_r$$

The threshold is used as a statistical rule for identifying unusually long durations within individual routes.

Chapter Summary

This chapter defined the main problems addressed by the project and established the technical objectives of the proposed solution. The raw ASG Airlines dataset contained several data quality issues, including missing airline information, inconsistent flight identifiers, unnecessary ingestion columns, duplicate records, and incorrect duration calculations for flights crossing midnight.

A structured data engineering approach was selected to address these issues. The pipeline uses Azure Databricks and PySpark for data processing and Delta Lake for maintaining the different stages of the data. Route-level statistical measures are used to identify unusually long flight durations, while Power BI is used to present the processed information through interactive dashboards.

The next chapter describes the dataset, technologies, tools, and methodology used to implement the proposed data engineering and analytics solution.

CHAPTER 3

DATASET AND TECHNOLOGIES

3.1 Dataset Description

The project uses a domestic airline flight dataset representing the operations of ASG Airlines. The dataset contains information about flights operated between different domestic airports and includes fields related to flight identification, airline, origin, destination, departure time, arrival time, and duration.

The dataset consists of **1,005 flight records** covering **four airlines** and **30 unique domestic routes**.

The major airports represented in the dataset include:

- Mumbai (BOM)
- Delhi (DEL)
- Kolkata (CCU)
- Bangalore (BLR)
- Chennai (MAA)
- Hyderabad (HYD)

The dataset was provided in **CSV format** and was used as the input for the data engineering pipeline developed in Azure Databricks.

3.1.1 Dataset Summary

Dataset Characteristic	Value
Total flight records	1,005
Number of airlines	4
Number of unique routes	30
Data format	CSV
Main analysis	Flight operations
Processing platform	Azure Databricks
Processing framework	PySpark

The four airlines represented in the dataset are:

Airline Prefix

IndiGo 6F

Airline Prefix

Air India AI

SpiceJet SJ

Vistara UK

3.2 Dataset Schema

The initial examination of the dataset was performed to understand the available fields and their data types.

3.2.1 Primary Dataset Fields

Field	Data Type	Description
flight_id	String	Identifier associated with a flight
airline	String	Name of the operating airline
source	String	IATA code of the origin airport
destination	String	IATA code of the destination airport
departure_time	String	Flight departure time in HH:MM format
arrival_time	String	Flight arrival time in HH:MM format
duration	Numeric/String	Flight duration available in the source data

The flight_id, source, and destination fields were used to identify individual flights and routes. The departure and arrival time fields were used to calculate and validate flight duration.

3.3 Initial Dataset Assessment

Before applying the transformation process, the raw dataset was examined to identify data quality issues.

The assessment identified the following conditions.

3.3.1 Unnecessary Columns

The raw dataset contained additional columns named _c7, _c8, _c9, and _c10. These columns did not contain useful flight information and were removed during data preparation.

3.3.2 Missing Airline Values

Some records contained missing or unknown values in the airline field.

Since airline-related prefixes were present in the flight_id field, these values were used to identify the airline wherever a valid prefix could be extracted.

3.3.3 Flight Identifier Patterns

The flight_id field contains airline-related prefixes. These prefixes were extracted during data processing using regular expressions.

The mapping used was:

Prefix Airline

6F IndiGo
AI Air India
SJ SpiceJet
UK Vistara

This information was used to improve the completeness of the airline field.

3.3.4 Time Representation

The departure and arrival times were provided in **HH:MM format** without an explicit date component.

This creates a special case for flights that cross midnight. For example, a flight departing at 23:15 and arriving at 01:45 would appear to have an arrival time smaller than its departure time.

Therefore, the time values were converted into minutes from midnight before calculating flight duration.

For a time represented by hour H and minute M :

$$T = 60H + M$$

The departure and arrival times are therefore represented as:

$$\begin{aligned} T_{dep} &= 60H_{dep} + M_{dep} \\ T_{arr} &= 60H_{arr} + M_{arr} \end{aligned}$$

The corrected flight duration is calculated as:

$$D = \begin{cases} T_{arr} - T_{dep}, & T_{arr} \geq T_{dep} \\ (T_{arr} + 1440) - T_{dep}, & T_{arr} < T_{dep} \end{cases}$$

where:

- D = flight duration in minutes
- T_{dep} = departure time in minutes
- T_{arr} = arrival time in minutes
- 1440= number of minutes in 24 hours

This calculation ensures that flights crossing midnight are assigned a positive duration.

3.4 Exploratory Data Analysis

An initial exploratory analysis was performed before the final dashboard development. The purpose of this analysis was to understand the distribution of flights across airlines and routes and to examine the duration values.

3.4.1 Airline-wise Flight Distribution

The 1,005 records are distributed across the four airlines as follows:

Airline Flight Records Percentage

IndiGo	273	27.16%
Air India	255	25.37%
SpiceJet	247	24.58%

Airline Flight Records Percentage

Vistara	230	22.89%
Total	1,005	100%

The distribution shows that the number of records is relatively balanced across the four airlines, with IndiGo having the highest number of records in the dataset.

3.4.2 Route Distribution

The dataset contains **30 unique domestic routes**. Some routes have a higher number of records than others.

The five routes with the highest number of records are:

Rank Route		Number of Flights Percentage	
1	BOM → CCU	90	8.96%
2	CCU → DEL	72	7.16%
3	MAA → BLR	65	6.47%
4	BLR → BOM	60	5.97%
5	HYD → MAA	57	5.67%
—	Remaining 25 routes	661	65.77%
Total 30 routes		1,005	100%

This distribution provides a basis for comparing flight activity between frequently used and less frequently used routes.

3.4.3 Flight Duration Distribution

The corrected flight duration values were examined to understand the overall range and variation in the dataset.

Duration Metric Value

Minimum duration	30 minutes
Maximum duration	300 minutes
Mean duration	164.62 minutes
Median duration	166 minutes
Standard deviation	42.18 minutes

The difference between the mean and median indicates that the overall duration distribution is relatively balanced, although variations exist between different routes.

Since different routes have different typical durations, route-level statistics were used for the anomaly analysis rather than applying a single threshold to the entire dataset.

3.5 Technologies Used

The project uses a combination of cloud-based data engineering, distributed processing, data storage, and business intelligence technologies.

The main technologies are:

Technology	Purpose
Azure Databricks	Data processing environment

Technology	Purpose
Apache Spark	Distributed data processing
PySpark	Data transformation and analysis
Delta Lake	Storage of processed data
Power BI Desktop	Dashboard and visualization
DAX	Creation of analytical measures
Python	Supporting data-processing logic

3.6 Azure Databricks

Azure Databricks was used as the primary environment for developing and executing the data processing pipeline.

The Databricks workspace was used to:

- Load the raw CSV dataset.
- Examine the dataset structure.
- Process the data using PySpark.
- Perform data cleaning and transformation.
- Create the Bronze, Silver, and Gold layers.
- Perform route-level statistical calculations.
- Prepare the final analytical data for Power BI.

Databricks provides a notebook-based environment in which the different stages of the data pipeline can be developed and tested.

3.7 Apache Spark and PySpark

Apache Spark is a distributed data processing framework. **PySpark** provides the Python interface for working with Spark.

PySpark was selected for this project because it provides functions for processing structured datasets and performing transformations efficiently.

The following PySpark operations were used during the project:

- DataFrame creation and manipulation.
- Column selection and renaming.
- Filtering and conditional transformations.
- Regular expression extraction.
- Duplicate removal.
- Time conversion.
- Aggregation.
- Window functions.
- Statistical calculations.

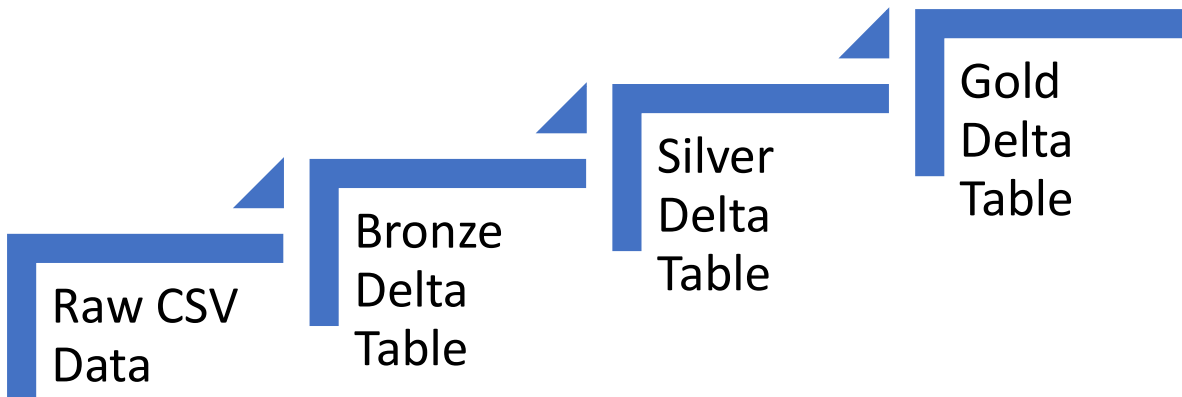
For example, Spark functions such as `regexp_extract()`, `when()`, and `otherwise()` were used to implement data-cleaning rules.

Window functions were also used to calculate route-level statistical values while retaining the individual flight records required for subsequent analysis.

3.8 Delta Lake

Delta Lake was used to store the processed datasets in the different stages of the pipeline.

The project follows the Bronze-Silver-Gold structure:



3.8.1 Bronze Layer

The Bronze layer contains the initially ingested data after basic structural processing.

The main activities include:

- Reading the raw CSV file.
- Examining the schema.
- Removing unnecessary ingestion columns.
- Saving the resulting data as a Delta table.

3.8.2 Silver Layer

The Silver layer contains the cleaned and transformed flight records.

The main transformations include:

- Airline identification.
- Missing-value handling.
- Duplicate removal.
- Time conversion.
- Cross-midnight duration correction.
- Route creation.
- Time-of-day classification.

3.8.3 Gold Layer

The Gold layer contains the data prepared for analysis and visualization.

It includes the cleaned flight information along with calculated analytical fields such as:

- Route mean duration.
- Route standard deviation.
- Route-specific threshold.
- Anomaly classification.

This layer serves as the main data source for the Power BI analysis.

3.9 Power BI

Microsoft Power BI Desktop was used to visualize the processed flight data and present the results through interactive dashboards.

The Gold-layer dataset was imported or connected to Power BI and used to create charts, KPI cards, filters, and analytical visuals.

The dashboard provides different views of the dataset, including:

- Overall flight activity.
- Airline-wise analysis.
- Flight duration analysis.
- Route traffic.
- Airport and hub analysis.
- Anomaly analysis.
- Key influencer analysis.

Interactive filters allow users to examine the data by airline, route, airport, and other available dimensions.

3.10 DAX

Data Analysis Expressions (DAX) was used to create calculated measures within Power BI.

Examples of measures used in the dashboard include:

Total Flights

$$Total\ Flights = COUNTROWS(FlightTable)$$

Average Flight Duration

$$Average\ Duration = \frac{\sum Duration}{Number\ of\ Flights}$$

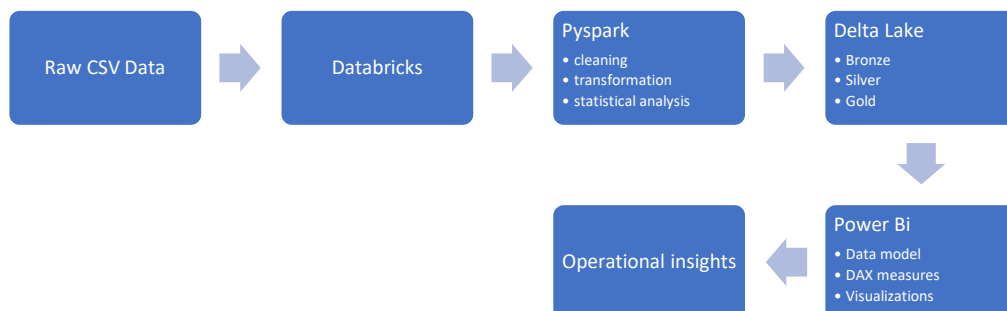
Anomaly Percentage

$$Anomaly\ Percentage = \frac{Number\ of\ Anomalous\ Flights}{Total\ Flights} \times 100$$

These measures allow the dashboard values to change dynamically when filters are applied.

3.11 Overall Technology Architecture

The technologies used in the project work together as a single data processing and reporting pipeline.



This architecture separates data processing from visualization and provides a structured path from the raw dataset to the final analytical dashboards.

Chapter Summary

This chapter described the dataset and technologies used in the ASG Airlines project. The dataset contains 1,005 domestic flight records covering four airlines and 30 unique routes.

The initial assessment identified issues such as unnecessary columns, missing airline values, embedded airline prefixes, and time values that required special handling for overnight flights.

Azure Databricks and PySpark were used for data processing and transformation, while Delta Lake was used to organize the data into Bronze, Silver, and Gold layers. Power BI and DAX were then used to create analytical measures and interactive dashboards from the curated data.

The next chapter presents the detailed implementation of the data engineering pipeline, including the data ingestion, cleaning, transformation, feature engineering, and statistical analysis steps.

Chapter 4

DATA PROCESSING AND PIPELINE IMPLEMENTATION

4.1 Introduction

This chapter describes the implementation of the data processing pipeline developed for the ASG Airlines case study. The raw flight dataset was processed using Azure Databricks and PySpark to identify data quality issues, standardize the records, correct flight-duration calculations, and prepare analytical datasets for Power BI.

The implementation follows a layered approach consisting of the **Bronze, Silver, and Gold layers**. The Bronze layer retains the ingested raw data, the Silver layer contains cleaned and standardized records, and the Gold layer contains business-ready information used for analysis and visualization.

The main processing activities include schema inspection, removal of unnecessary columns, handling of missing airline information, flight identifier processing, duplicate detection, cross-midnight duration correction, feature engineering, and route-level anomaly detection.

4.2 Data Pipeline Architecture

The implemented pipeline follows the sequence:

Raw CSV → Bronze Layer → Silver Layer → Gold Layer → Power BI

4.2.1 Bronze Layer

The Bronze layer contains the raw flight dataset as received. At this stage, the data is preserved before applying major transformations.

The purpose of this layer is to provide a reference point for the original dataset and allow the subsequent processing stages to be traced back to the source data.

4.2.2 Silver Layer

The Silver layer contains the cleaned and standardized flight records. The major transformations performed at this stage include:

- Removing unnecessary ingestion columns.
- Standardizing column names and data types.
- Identifying missing or unknown airline values.
- Extracting airline information from flight identifiers.
- Removing duplicate records.
- Converting departure and arrival times into usable time values.
- Correcting flight durations for flights crossing midnight.
- Creating additional analytical fields.

4.2.3 Gold Layer

The Gold layer contains the final datasets prepared for business analysis. It includes derived fields and aggregated information required for Power BI reporting.

The Gold layer supports analysis of:

- Total flights.
- Airline distribution.
- Route traffic.
- Flight duration.
- Overnight flights.
- Route-level duration anomalies.
- Airport and route patterns.

4.3 Raw Data Ingestion

The original flight dataset was provided in CSV format and uploaded into the Azure Databricks environment.

The first step was to read the dataset using PySpark and inspect its structure.

The following checks were performed during ingestion:

- Number of records.
- Number of columns.
- Column names.
- Data types.
- Missing values.
- Duplicate records.
- Unique airline values.
- Unique routes.
- Departure and arrival time formats.

The initial dataset contained **1,005 flight records**. The preliminary inspection also identified additional columns generated during the ingestion process that did not contain useful business information.

4.4 Schema Inspection and Data Type Validation

After ingestion, the schema of the dataset was examined using PySpark.

The schema inspection was used to determine whether the columns had appropriate data types for subsequent processing. In particular, the departure time, arrival time, and duration-related fields required attention because time values cannot be reliably processed as ordinary strings.

The following categories of fields were identified:

Field Category	Examples	Processing Requirement
Flight identification	flight_id	Standardization and airline extraction
Airline	airline	Missing-value handling
Route	source, destination	Route construction
Time	departure_time, arrival_time	Time conversion

Field Category	Examples	Processing Requirement
Duration	duration	Validation and correction
Ingestion artifacts	_c7-_c10	Removal

Schema validation was performed before applying transformations so that later calculations could be carried out consistently.

4.5 Removal of Unnecessary Columns

The raw dataset contained additional columns named _c7, _c8, _c9, and _c10. These columns were identified as unnecessary for the flight analysis.

They were removed during the cleaning process to avoid carrying irrelevant fields into the analytical layers.

This step simplified the dataset and reduced the possibility of including ingestion-related artifacts in the Power BI model.

4.6 Handling Missing and Unknown Airline Information

Some flight records did not contain a valid airline name. In addition, certain records contained "UNKNOWN" values.

However, the airline information could be derived from the prefix present in the flight_id field.

The following mapping was applied:

Flight Prefix Airline

6F	IndiGo
AI	Air India
SJ	SpiceJet
UK	Vistara

The airline field was therefore reconstructed using the flight identifier wherever the original airline value was missing or marked as unknown.

This approach helped maintain a consistent airline classification across the dataset.

4.7 Flight Identifier Processing

The flight_id field was examined to identify airline prefixes and maintain consistent flight identification.

For example, a flight identifier beginning with 6F was associated with IndiGo, while an identifier beginning with AI was associated with Air India.

Regular-expression-based extraction was used to identify the relevant prefix from the flight identifier.

This processing allowed airline information to be recovered without requiring an additional external data source.

4.8 Duplicate Record Detection

Duplicate records were examined as part of the data quality process.

Duplicate entries can affect flight counts, route statistics, and airline-level comparisons. Therefore, duplicate records were identified using relevant flight-level fields and removed where appropriate.

The cleaned dataset was then used for further calculations and analytical processing.

4.9 Route Construction

A route field was created by combining the source and destination airport codes.

For a flight travelling from Mumbai to Kolkata, the route was represented as:

BOM-CCU

This representation simplified route-level grouping and analysis.

The processed dataset contained **30 unique domestic routes**.

The route field was later used for:

- Flight volume analysis.
- Average duration calculation.
- Route-level statistical analysis.
- Anomaly detection.
- Power BI visualizations.

4.10 Cross-Midnight Flight Duration Correction

One of the important data quality issues identified in the dataset was the incorrect calculation of duration for flights that crossed midnight.

For example, consider a flight departing at **23:15** and arriving at **01:45** the following day.

If the times are treated as simple clock values, the calculation would incorrectly produce a negative duration.

To solve this problem, departure and arrival times were converted into minutes from midnight.

For departure time:

$$T_{dep} = 60H_{dep} + M_{dep}$$

For arrival time:

$$T_{arr} = 60H_{arr} + M_{arr}$$

The duration was then calculated as:

$$D = \begin{cases} T_{arr} - T_{dep}, & T_{arr} \geq T_{dep} \\ (T_{arr} + 1440) - T_{dep}, & T_{arr} < T_{dep} \end{cases}$$

where **1440 represents the number of minutes in 24 hours**.

For the example:

$$\begin{aligned} T_{dep} &= 23(60) + 15 = 1395 \\ T_{arr} &= 1(60) + 45 = 105 \end{aligned}$$

Since the arrival time is smaller than the departure time:

$$\begin{aligned} D &= (105 + 1440) - 1395 \\ D &= 150 \text{ minutes} \end{aligned}$$

Therefore, the actual flight duration is **150 minutes**.

This correction prevents overnight flights from being incorrectly classified as negative or unusually short-duration flights.

4.11 Feature Engineering

Additional fields were created from the cleaned flight data to support analysis.

The main derived features include:

4.11.1 Route

The source and destination airport codes were combined to create a route identifier.

4.11.2 Duration in Minutes

The corrected duration was stored in minutes to simplify numerical analysis and visualization.

4.11.3 Overnight Flight Indicator

An indicator was created to identify flights where the arrival time occurred after midnight relative to the departure time.

This field was used to distinguish overnight flights from regular daytime flights.

4.11.4 Departure and Arrival Components

Departure and arrival time information was converted into usable components for further analysis.

These fields allow the dataset to be grouped and compared according to flight timing.

4.12 Route-Level Statistical Analysis

Flight duration varies between routes because of differences in distance, airport location, flight paths, and operating conditions.

Therefore, a single duration threshold was not applied to every route. Instead, route-specific statistics were calculated.

For each route r , the mean duration was calculated as:

$$\mu_r = \frac{1}{n_r} \sum_{i=1}^{n_r} D_i$$

where:

- n_r = number of flights on route r
- D_i = duration of flight i
- μ_r = mean duration for route r

The standard deviation was calculated as:

$$\sigma_r = \sqrt{\frac{1}{n_r} \sum_{i=1}^{n_r} (D_i - \mu_r)^2}$$

A route-specific threshold was then defined as:

$$Threshold_r = \mu_r + 1.5\sigma_r$$

A flight was marked as a duration anomaly when:

$$D_i > Threshold_r$$

Using route-specific thresholds makes the analysis more meaningful because a duration that is unusual for a short route may be normal for a longer route.

4.13 Creation of the Gold Analytical Dataset

After completing the cleaning and transformation steps, the processed records were organized into a business-ready dataset.

The Gold dataset contains the fields required for analytical reporting, including:

- Flight identifier.
- Airline.
- Source airport.
- Destination airport.
- Route.
- Departure time.
- Arrival time.

- Corrected duration.
- Overnight indicator.
- Route-level statistics.
- Anomaly indicator.

This dataset was subsequently made available for Power BI analysis.

4.14 Data Quality Validation

Validation checks were performed after the transformations to ensure that the processed data was suitable for analysis.

The main validation checks included:

Validation Check	Purpose
Record count	Verify that processing did not unintentionally remove records
Null-value check	Identify remaining missing values
Duplicate check	Ensure duplicate records were handled
Airline validation	Confirm valid airline classifications
Route validation	Confirm source-destination combinations
Duration validation	Identify invalid or unrealistic durations
Overnight validation	Verify cross-midnight flight identification
Anomaly validation	Confirm route-level threshold calculations

These checks helped confirm that the cleaned dataset was consistent before being transferred to the reporting layer.

4.15 Integration with Power BI

The Gold dataset was used as the primary analytical source for Power BI.

The processed data was loaded into Power BI Desktop, where measures and calculated fields were created using DAX.

The Power BI model was then used to develop multiple dashboard pages covering:

- Overall flight operations.
- Airline performance.
- Route traffic.
- Flight duration.
- Overnight flights.
- Duration anomalies.
- Route-level patterns.

The use of the Gold layer ensures that the Power BI reports are based on processed and standardized data rather than the original raw dataset.

Chapter Summary

The implementation described in this chapter transformed the raw ASG Airlines flight dataset into a structured analytical dataset.

The processing pipeline addressed the major data quality problems identified during the initial assessment, including unnecessary ingestion columns, missing airline information, inconsistent identifiers, duplicate records, and incorrect duration calculations for flights crossing midnight.

Route construction, feature engineering, and route-level statistical analysis were then applied to prepare the data for business reporting. The resulting Gold dataset was connected to Power BI and used as the foundation for the dashboard and analytical visualizations discussed in the following chapter.

Chapter 5

POWER BI DASHBOARD DEVELOPMENT AND DATA ANALYSIS

5.1 Introduction

This chapter presents the development of the Power BI dashboard using the cleaned and transformed flight dataset generated from the data processing pipeline. The purpose of the dashboard is to convert the processed flight records into meaningful visual information that can be used to understand airline distribution, route traffic, flight duration, overnight operations, and duration anomalies.

The Gold-layer dataset was used as the primary source for Power BI. DAX measures were created to calculate important operational metrics, while different visualizations were used to present the results in an understandable form.

The dashboard was organized into multiple pages so that overall performance, route-level patterns, duration behaviour, and anomalies could be examined separately.

5.2 Power BI Data Model

The processed Gold dataset was imported into Power BI Desktop for reporting and analysis.

The primary table used in the Power BI model was:

Airlines_Cleaned_PowerBI

The table contains the cleaned flight records and the derived fields generated during the data-processing stage.

Important fields used in the Power BI model include:

Field	Purpose
flight_id	Identifies the flight
airline	Identifies the operating airline
source	Departure airport
destination	Arrival airport
route	Combined source-destination route
departure_time	Scheduled departure time
arrival_time	Scheduled arrival time
duration	Corrected flight duration
overnight	Identifies overnight flights
anomaly	Identifies unusual route-level durations

The cleaned structure allowed Power BI to perform aggregations without repeatedly applying the data-cleaning operations performed in Databricks.

5.3 Development of DAX Measures

DAX was used to create measures required for the dashboard.

5.3.1 Total Flights

The total number of flight records was calculated using:

Total Flights = COUNTROWS('Airlines_Cleaned_PowerBI')

This measure provides the overall number of flight records available for analysis.

5.3.2 Average Flight Duration

The average flight duration was calculated using:

Average Duration = AVERAGE('Airlines_Cleaned_PowerBI'[duration])

This measure provides the average duration of the flights represented in the selected filter context.

5.3.3 Overnight Flights

The number of overnight flights was calculated by filtering the overnight indicator.

Overnight Flights =

```
CALCULATE(  
    [Total Flights],  
    'Airlines_Cleaned_PowerBI'[overnight] = 1  
)
```

5.3.4 Anomalous Flights

The number of flights identified as duration anomalies was calculated as:

Anomalous Flights =

```
CALCULATE(  
    [Total Flights],  
    'Airlines_Cleaned_PowerBI'[anomaly] = 1  
)
```

These measures were used in KPI cards and other dashboard visualizations.

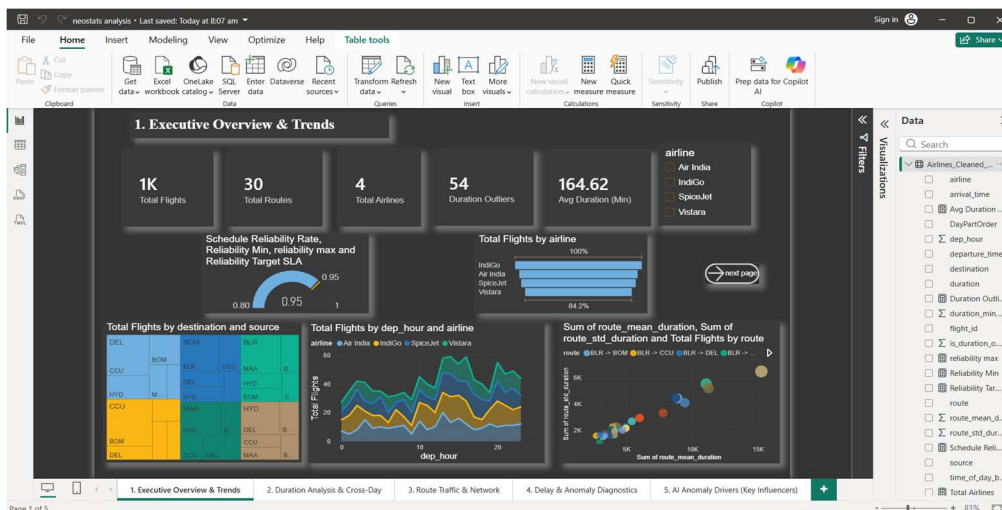
5.4 Dashboard Page 1 – Executive Overview

The first dashboard page provides a high-level summary of the flight dataset.

The main KPI cards present:

- Total Flights
- Number of Airlines
- Number of Routes
- Average Flight Duration
- Overnight Flights
- Anomalous Flights

The page also provides a general comparison of airline activity and route distribution.



5.4.1 Airline Distribution

A chart was used to compare the number of flights operated by each airline.

Based on the processed dataset:

Airline Flights Percentage

IndiGo	273	27.16%
Air India	255	25.37%
SpiceJet	247	24.58%
Vistara	230	22.89%
Total	1,005	100%

IndiGo has the highest number of records in the dataset, followed by Air India, SpiceJet, and Vistara. The distribution is relatively balanced across the four airlines.

5.5 Dashboard Page 2 – Flight Duration Analysis

The second dashboard page focuses on flight duration.

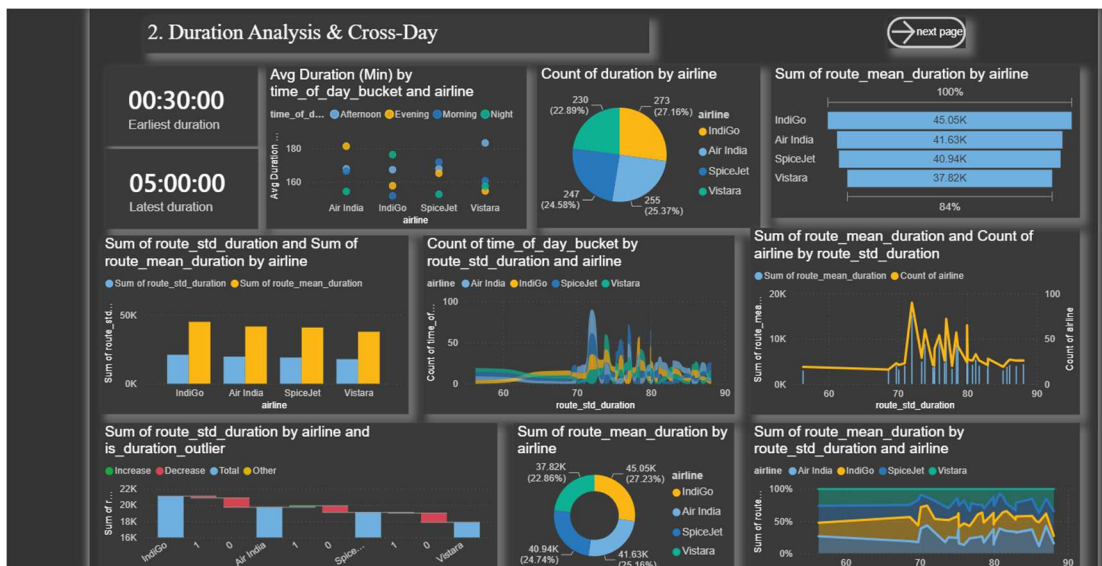
The analysis uses the corrected duration generated during the data-processing stage. This is important because flights crossing midnight require special handling to avoid incorrect duration values.

The duration analysis includes:

- Average duration.
- Minimum duration.
- Maximum duration.
- Duration distribution.
- Airline-wise duration comparison.
- Route-wise duration comparison.

The dataset contains flight durations ranging from approximately **30 minutes to 300 minutes**.

The overall average duration can also be compared across airlines and routes to identify differences in operational patterns.



5.6 Dashboard Page 3 – Route Traffic and Network Analysis

The third dashboard page focuses on route-level flight activity.

A route represents the movement between a source and destination airport.

The dataset contains **30 unique routes**. The most frequently represented routes include:

Route	Flights	Percentage
BOM → CCU	90	8.96%
CCU → DEL	72	7.16%
MAA → BLR	65	6.47%
BLR → BOM	60	5.97%
HYD → MAA	57	5.67%

The remaining 25 routes account for 661 records.

A bar chart was used to compare route volumes, while route-based visuals were used to examine the distribution of flights between major airports.

This analysis helps identify routes with relatively high levels of activity within the dataset.

5.7 Airport-Level Analysis

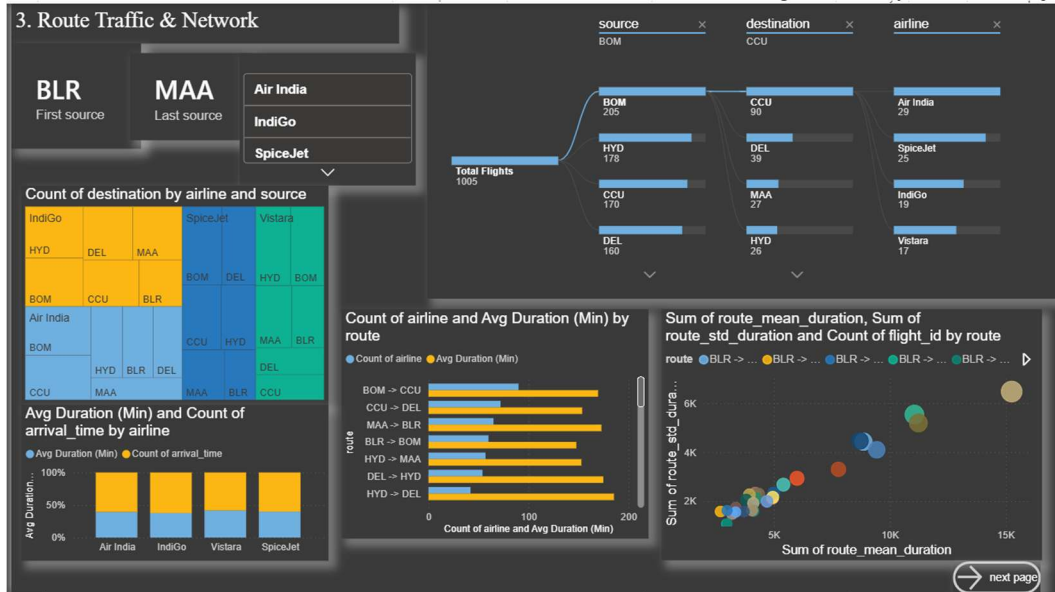
The source and destination airport fields were also used to examine airport activity.

The major airport codes represented in the dataset include:

- BOM – Mumbai
- DEL – Delhi
- CCU – Kolkata
- BLR – Bangalore
- MAA – Chennai
- HYD – Hyderabad

Airport-level analysis helps identify locations that appear frequently as departure or arrival points.

The results can also be used to understand the concentration of flight activity around major airports.



5.8 Dashboard Page 4 – Overnight Flight and Anomaly Analysis

The fourth dashboard page focuses on flights operating across midnight and flights with unusual duration values.

5.8.1 Overnight Flight Analysis

An overnight flight is identified when the arrival time occurs after midnight relative to the departure time.

The processed dataset contains **122 overnight flights**.

The overnight indicator allows the dashboard to compare:

- Overnight versus non-overnight flights.
- Overnight flights by airline.
- Overnight flights by route.
- Distribution of overnight operations.

This provides a clearer view of flight timing patterns.

5.8.2 Duration Anomaly Analysis

Duration anomalies were identified using route-level statistical thresholds.

For each route, the mean and standard deviation of flight duration were calculated. A route-specific threshold was then established using:

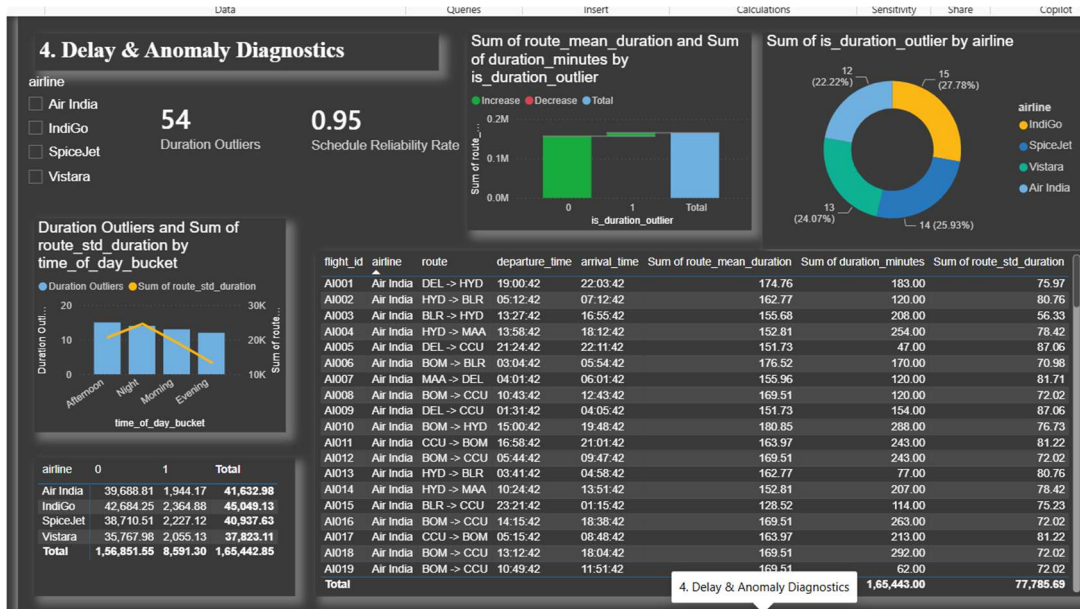
$$Threshold_r = \mu_r + 1.5\sigma_r$$

Flights exceeding their respective route threshold were marked as anomalies.

The dashboard allows these records to be examined by:

- Route.
- Airline.
- Duration.
- Expected route-level duration.
- Difference from the route threshold.

This provides a more meaningful comparison than applying one fixed duration threshold to every route.



5.9 Dashboard Page 5 – Key Influencer Analysis

The final dashboard page uses Power BI's Key Influencers analysis to examine factors associated with selected flight characteristics.

The analysis can be used to investigate how fields such as:

- Airline
- Route
- Source airport
- Destination airport
- Overnight status

are associated with selected analytical outcomes.

For example, the analysis can help identify whether certain airlines or routes appear more frequently among overnight flights or duration anomalies.

The purpose of this page is exploratory rather than predictive. It helps users examine relationships within the available flight records.

5.10 Interactive Filtering

Power BI slicers and filters were incorporated to allow users to explore the dataset interactively.

The main filtering dimensions include:

- Airline.
- Route.
- Source airport.
- Destination airport.
- Overnight status.
- Anomaly status.

When a filter is applied, the KPI cards and visualizations update according to the selected records.

For example, selecting a particular airline allows its flight count, average duration, route distribution, overnight flights, and anomaly records to be examined without creating a separate report.

5.11 Key Analytical Findings

The dashboard provides several observations from the processed dataset.

5.11.1 Airline Distribution

The four airlines have relatively similar representation in the dataset. IndiGo has the highest number of records with 273 flights, while Vistara has the lowest with 230.

5.11.2 Route Concentration

Flight activity is not evenly distributed across all routes. A small number of routes account for a noticeable proportion of the records, with BOM–CCU being the most frequently represented route.

5.11.3 Flight Duration

Flight durations vary considerably across routes. Therefore, comparing durations using route-specific statistics provides a more useful basis for anomaly detection than using one global threshold.

5.11.4 Overnight Operations

The dataset contains 122 overnight flights. The overnight indicator allows these flights to be separated from normal daytime operations and examined by airline and route.

5.11.5 Duration Anomalies

The route-level anomaly analysis identifies flights whose duration is unusually high compared with other flights operating on the same route. These records can be further investigated to determine possible operational or data-quality reasons.

5.12 Business Value of the Dashboard

The Power BI dashboard converts the processed flight data into a format that is easier for users to interpret and explore.

The dashboard supports:

- Monitoring the distribution of flights across airlines.
- Identifying frequently used routes.
- Comparing flight durations.
- Examining overnight operations.
- Detecting unusual duration records.
- Filtering information by airline and route.
- Supporting further investigation of operational patterns.

Instead of manually examining individual flight records, users can use the dashboard to identify important patterns through interactive visualizations.

Chapter Summary

This chapter presented the development of the Power BI reporting layer for the ASG Airlines case study.

The cleaned Gold dataset was imported into Power BI and used to create DAX measures, KPI cards, charts, route analysis, duration analysis, overnight-flight analysis, and anomaly analysis.

The dashboard provides an interactive view of the flight dataset and allows users to examine airline, route, timing, duration, and anomaly patterns. The results obtained from these analyses are discussed further in the following chapter.



Chapter 6

CHAPTER 6

RESULTS, FINDINGS AND DISCUSSION

6.1 Introduction

This chapter presents the results obtained after implementing the data engineering pipeline and Power BI reporting solution for the ASG Airlines case study. The results are based on the cleaned and transformed flight dataset prepared using Azure Databricks and PySpark.

The analysis focuses on data quality improvement, airline distribution, route activity, flight duration, overnight operations, and duration anomalies. The Power BI dashboard was used to explore these aspects through interactive visualizations and calculated measures.

The findings presented in this chapter demonstrate how the processed dataset can be used to obtain useful operational insights from flight records.

6.2 Results of Data Processing

The first stage of evaluation focused on the quality of the data after processing.

The raw dataset contained several issues that could affect analysis if they were not addressed. These included unnecessary ingestion columns, missing airline information, inconsistent flight identifiers, duplicate records, and incorrect duration calculations for flights crossing midnight.

The implemented pipeline addressed these issues through a sequence of cleaning and transformation operations.

Table 6.1 Data Processing Results

Data Issue	Processing Performed	Result
Unnecessary columns	Removed _c7-_c10	Cleaner analytical schema
Missing airline values	Airline prefix extraction	Airline information reconstructed
Unknown airline values	Identifier-based mapping	Standardized airline classification
Duplicate records	Duplicate detection and removal	Reduced repeated records

Data Issue	Processing Performed	Result
Cross-midnight durations	Time-based correction	Correct duration calculation
Inconsistent route representation	Route construction	Standard route field
Unusual durations	Route-level statistical analysis	Anomaly identification

The processing stage therefore provided a more consistent dataset for subsequent analysis.

6.3 Overall Dataset Results

After processing, the dataset contained **1,005 flight records**, representing **4 airlines** and **30 unique routes**.

The main descriptive characteristics of the dataset are summarized below.

Table 6.2 Overall Dataset Summary

Metric	Value
Total flight records	1,005
Airlines	4
Unique routes	30
Minimum duration	30 minutes
Maximum duration	300 minutes
Average duration	164.62 minutes
Median duration	166 minutes
Standard deviation	42.18 minutes
Overnight flights	122

These values provide an overall understanding of the dataset before examining individual airlines and routes.

6.4 Airline-Level Findings

The airline distribution shows that all four airlines have substantial representation in the dataset.

Table 6.3 Airline Distribution

Airline	Flights	Percentage
IndiGo	273	27.16%
Air India	255	25.37%
SpiceJet	247	24.58%
Vistara	230	22.89%
Total	1,005	100%

IndiGo has the highest number of records with 273 flights, accounting for 27.16% of the dataset. Air India follows with 255 flights, while SpiceJet and Vistara account for 247 and 230 flights respectively.

The difference between the highest and lowest airline counts is relatively small. Therefore, the dataset does not show an extremely uneven representation of the four airlines.

This balanced distribution is useful for comparing airline-level patterns without one airline dominating the dataset.

6.5 Route-Level Findings

Route analysis showed that the 1,005 flight records were distributed across 30 unique domestic routes.

The five most frequently represented routes were:

Table 6.4 Top Routes

Rank	Route	Flights	Percentage
1	BOM → CCU	90	8.96%
2	CCU → DEL	72	7.16%
3	MAA → BLR	65	6.47%
4	BLR → BOM	60	5.97%
5	HYD → MAA	57	5.67%

The BOM–CCU route has the highest representation with 90 flights.

The top five routes together account for **344 records**, while the remaining 25 routes account for **661 records**.

This indicates that flight activity is distributed across a relatively broad set of routes rather than being concentrated entirely within the top few routes.

6.6 Flight Duration Findings

Flight duration was analyzed after correcting the time calculation for flights crossing midnight.

The observed duration range was approximately **30 to 300 minutes**, with an average duration of **164.62 minutes** and a median of **166 minutes**.

The relatively close values of the mean and median indicate that the overall duration distribution is not strongly affected by extremely large values.

However, duration varies between routes. A duration that may be considered high for one route may be normal for another route.

For this reason, route-specific analysis was used instead of applying a single global duration threshold.

6.7 Cross-Midnight Duration Correction Results

The cross-midnight correction was an important part of the data-processing stage.

Without the correction, a flight departing late at night and arriving after midnight could produce an incorrect negative duration.

For example:

Departure: 23:15

Arrival: 01:45

The corrected duration is:

$$(01:45 + 24:00) - 23:15 = 2:30$$

Therefore:

$$\text{Duration} = 150 \text{ minutes}$$

The implementation ensured that such flights were represented using their actual elapsed duration rather than a negative or incorrect value.

This improved the reliability of subsequent duration statistics and anomaly detection.

6.8 Overnight Flight Findings

The analysis identified **122 overnight flights** in the dataset.

The overnight indicator created during the data-processing stage allowed these records to be separated from other flights.

The dashboard was used to examine overnight flights by:

- Airline.
- Route.
- Departure and arrival timing.
- Flight frequency.

This analysis provides an additional perspective on the operating schedule represented in the dataset.

It also demonstrates the usefulness of derived features in transforming basic flight records into more informative analytical variables.

6.9 Route-Level Anomaly Findings

Anomaly detection was performed using route-specific duration statistics.

For each route, the mean duration and standard deviation were calculated.

The threshold was defined as:

$$Threshold_r = \mu_r + 1.5\sigma_r$$

A flight was classified as a duration anomaly when:

$$D_i > Threshold_r$$

This method allows each route to have its own expected duration range.

For example, a duration of 250 minutes may be unusual on a relatively short route but may not be unusual on a longer route. Therefore, route-specific thresholds provide a more appropriate basis for identifying unusual records.

The anomalies identified by this method should be treated as **records requiring further investigation**, rather than automatically being considered operational failures or delays.

6.10 Data Quality Validation Results

The final dataset was checked after the major transformations were completed.

The validation focused on:

- Record count.
- Missing values.
- Airline classification.
- Duplicate records.
- Route consistency.
- Flight duration.
- Overnight classification.
- Anomaly identification.

Table 6.5 Validation Summary

Validation Metric Result

Total Records 1,005

Unknown Airlines 69

Overnight Flights 122

Validation Metric Result

Duration Issues 1

The validation results indicate that the main data-processing operations were successfully applied. The remaining values should be interpreted according to the final cleaned dataset and the specific validation logic used in Databricks.

Note: If the final Silver/Gold dataset has changed after these figures were generated, these values should be updated before submission.

6.11 Power BI Dashboard Findings

The Power BI dashboard provided an interactive method for examining the processed data.

The dashboard made it possible to move from overall summary information to individual airline, route, and flight-level analysis.

The main findings available through the dashboard include:

1. **Airline activity** – IndiGo has the largest number of flight records in the dataset.
2. **Route activity** – BOM–CCU is the most frequently represented route.
3. **Duration variation** – Flight duration differs considerably between routes.
4. **Overnight operations** – 122 records are classified as overnight flights.
5. **Duration anomalies** – Route-level thresholds help identify unusually long flight-duration records.
6. **Interactive analysis** – Filters allow the results to be examined for specific airlines, routes, and flight categories.

6.12 Discussion of Results

The results demonstrate the importance of data preparation before business intelligence analysis.

If the raw dataset had been directly connected to Power BI, issues such as missing airline information, unnecessary columns, and incorrect cross-midnight durations could have affected the resulting analysis.

The implementation of the Bronze–Silver–Gold pipeline provided a structured method for separating raw data from cleaned and analytical data.

The route-specific anomaly approach was particularly useful because flight durations naturally differ across routes. Instead of treating every flight using the same threshold, the analysis considers the characteristics of the route on which the flight operates.

The Power BI dashboard then provided a convenient interface for examining these results without requiring users to work directly with the underlying PySpark transformations.

6.14 Limitations of the Results

The results should be interpreted within the characteristics of the available dataset.

The analysis is based on the flight records available in the project dataset and therefore represents the patterns contained within those records. The results should not be interpreted as a complete representation of the operations of the actual airlines.

Similarly, a duration anomaly indicates that a record has an unusually high duration relative to other records on the same route. It does not by itself establish the reason for the unusual duration.

Further analysis using additional operational information such as actual departure time, actual arrival time, weather conditions, aircraft information, or airport congestion data could provide additional context.

Chapter Summary

This chapter presented the results obtained from the implemented data engineering and business intelligence pipeline.

The processed dataset contained 1,005 flight records across four airlines and 30 routes. Airline distribution, route activity, flight duration, overnight operations, and route-level duration anomalies were analyzed using Power BI.

The results demonstrate that data cleaning and feature engineering significantly improve the usefulness of the original flight records for analytical purposes. The dashboard provides an interactive way to explore the resulting information and identify patterns that would be difficult to observe directly from the raw CSV data.

The following chapter presents the overall conclusion of the project and discusses possible future improvements.

Chapter 7

CONCLUSION AND FUTURE ENHANCEMENTS

7.1 Conclusion

The ASG Airlines project developed an end-to-end data engineering and analytics workflow for processing domestic flight records.

The pipeline used Azure Databricks and PySpark to clean the raw dataset, remove unnecessary fields, recover missing airline information, correct cross-midnight flight durations, create analytical features, and identify unusual route-level duration values.

The processed dataset was then used in Power BI to develop an interactive five-page dashboard covering airline distribution, route activity, flight duration, overnight operations, and anomalies.

The analysis of 1,005 flight records identified four airlines, 30 routes, 122 overnight flights, and 54 duration anomalies. The results demonstrate how structured data processing can improve the quality and usability of operational flight data for analytical reporting.

7.2 Future Enhancements

The project can be extended in several ways:

- Integrate real-time flight data using streaming technologies.
- Incorporate weather and airport information for deeper analysis.
- Develop machine learning models for flight-delay prediction.
- Use Azure Data Factory for scheduled pipeline execution.
- Add automated data-quality monitoring.
- Extend the Power BI dashboard with additional operational metrics.

These enhancements could provide more advanced predictive and real-time capabilities while building on the existing data engineering pipeline